

Natural Radionuclide Activity and Radiological Risk Assessment in Brazilian Coffee Samples



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1. INTRODUCTION

Brazil is the world's largest coffee producer, responsible for about 36.8% of global production in 2024. Besides economic importance, it is essential to ensure the radiological safety of coffee for consumers and international markets. Natural radionuclides such as ^{40}K , ^{226}Ra , and ^{228}Ra are present in soils and may be absorbed by plants. Due to the Goiânia accident in 1987, the artificial radionuclide ^{137}Cs must also be monitored. This study quantified radionuclide activity concentrations in Brazilian coffee, estimated the effective dose and lifetime cancer risk, and compared results with international reference limits.

2. MATERIALS AND METHODS

Seventeen ground coffee samples from Rio de Janeiro and Minas Gerais were analyzed by high-resolution gamma spectrometry using a High-Purity Germanium (HPGe) detector. Samples were sealed in polypropylene containers and stored for 45 days to establish secular equilibrium. Efficiency calibration was performed with the LabSOCS software. Activity concentrations of ^{40}K , ^{226}Ra , ^{228}Ra , and ^{137}Cs were determined from characteristic gamma emissions. The annual effective dose was calculated using ICRP 119 ingestion coefficients and IBGE consumption data. Cancer risk was estimated using USEPA risk factors.

3. RESULTS

The average activity concentrations were $562 \text{ Bq}\cdot\text{kg}^{-1}$ for ^{40}K , $1.1 \text{ Bq}\cdot\text{kg}^{-1}$ for ^{226}Ra , and $2.4 \text{ Bq}\cdot\text{kg}^{-1}$ for ^{228}Ra , while ^{137}Cs was below the detection limit in all samples. The annual effective dose ranged from 0.107 to $0.166 \text{ mSv}\cdot\text{y}^{-1}$, with an average of $0.133 \text{ mSv}\cdot\text{y}^{-1}$. The ^{40}K was the main contributor (88.9%), followed by ^{228}Ra (10.3%) and ^{226}Ra (0.8%).

Table 01: Distribution of the Effective dose in Brazilian coffee.

Nuclides	Contribution	Average (mSv.y-1)	Maximum (mSv.y-1)	Minimum (mSv.y-1)
^{40}K	88.9%	0.118	0.131	0.106
^{226}Ra	0.8%	0.001	0.025	0.000
^{228}Ra	10.3%	0.013	0.045	0.000

The calculated cancer risk was of the order of 10^{-6} , considered negligible. All values were below the UNSCEAR reference value ($290 \mu\text{Sv}\cdot\text{y}^{-1}$) and the ICRP limit ($1 \text{ mSv}\cdot\text{y}^{-1}$). When compared with other countries, the results were higher than Malaysia, lower than Saudi Arabia, and close to Iraq.

Table 02: Comparison the annual effective dose in coffee powder.

	Average of annual effective dose (mSv.y-1)
Present study	0.133
Khandaker (2020)	0.022
Alharbi (2017)	0.241
Al-Alawy (2020)	0.077

4. CONCLUSIONS

Brazilian coffee contains mainly ^{40}K , with lower concentrations of ^{226}Ra and ^{228}Ra , and no detectable ^{137}Cs . The estimated effective dose and cancer risk are below international safety limits, confirming that coffee consumption does not represent a radiological risk to the Brazilian population. This work provides baseline values for food safety assessments and contributes to future radiological monitoring of agricultural products in Brazil.

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